Supplementary Material

Road Extraction at the 5.8 m Limit. A LISS-IV Benchmark with Two-Tier, Human-Grounded Evaluation

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This document accompanies the ICVGIP 2026 paper. Section S1 defines the training targets and losses. Section S2 describes the strict APLS evaluator and reports it at all scales, directions and buffer sizes. Section S3 gives the stratified, human-traced, localisation, compute, leave-one-scene-out and seasonal results. Section S4 reports the pre-registered decoder study and the two training changes. Every number comes from the same result files as the paper, and the code is at https://github.com/Destroyer1543/liss-toporoadnet.

S1 TARGETS AND LOSSES

S1.1 Targets

Let y be the binary road mask of a tile. The centreline target s is the morphological skeleton of y. The width target is the Euclidean distance from each skeleton pixel to the nearest background pixel, so it is a half-width in pixels and is defined on skeleton pixels only. The affinity target has eight channels. With the offsets d_k taken in the order E, NE, N, NW, W, SW, S, SE, channel k is one at pixel p when both p and $p - d_k$ are skeleton pixels. The deep-supervision target is y max-pooled over 4×4 blocks, so that one-pixel roads survive at 1/4 scale.

S1.2 Outputs

The network predicts a mask logit ℓ_m, a centreline probability P_s, a half-width $W = \mathrm{softplus}(\cdot) + 0.5$, eight affinity logits and, from the 1/4-scale decoder features, a deep-supervision logit. The reconstruction branch rebuilds the mask as

$$M_{\mathrm{rec}}(p) = \max_{q \in N_9(p)} P_s(q)\, \exp\!\left(-\frac{\|p-q\|^2}{2\,W(q)^2}\right),$$

where $N_9(p)$ is the 9×9 neighbourhood of p. The fused logit is $\ell_f = \alpha\,\ell_m + \beta\,\mathrm{logit}(M_{\mathrm{rec}})$ with learned scalars α and β, and the road probability is $p_f = \sigma(\ell_f)$.

S1.3 Losses

All terms are per-pixel or per-sample means. With the focal Tversky loss $\mathrm{FT}(p, y) = \max(1 - \mathrm{TI}, 10^{-4})^{\gamma}$, where $\mathrm{TI} = \mathrm{TP}/(\mathrm{TP} + \alpha_t\,\mathrm{FN} + \beta_t\,\mathrm{FP})$, $\alpha_t = 0.7$, $\beta_t = 0.3$ and $\gamma = 0.75$, the total loss is

$$\begin{aligned}\mathcal{L} = {} & \mathcal{L}_{\mathrm{seg}}(p_f, y) + 0.4\,\mathcal{L}_{\mathrm{seg}}(p_4, y_4) \\ & + \mathbb{1}[e \geq 5]\,\big(1.0\,\mathcal{L}_{\mathrm{clDice}}(p_f, y) + 1.0\,\mathcal{L}_{\mathrm{skel}} \\ & \quad + 0.2\,\mathcal{L}_{\mathrm{width}} + 0.25\,\mathcal{L}_{\mathrm{aff}} + 0.1\,\mathcal{L}_{\mathrm{cons}}\big),\end{aligned}$$

with $\mathcal{L}_{\mathrm{seg}}(p, y) = \mathrm{FT}(p, y) + 0.5\,\mathrm{Dice}(p, y)$, p_4 and y_4 the deep-supervision output and target, and e the epoch. $\mathcal{L}_{\mathrm{clDice}}$ is soft clDice with 10 soft skeleton iterations. $\mathcal{L}_{\mathrm{skel}}$ is binary cross-entropy on the centreline logit with positive weight 10. $\mathcal{L}_{\mathrm{width}}$ is a Huber loss between W and the width target on ground-truth skeleton pixels. $\mathcal{L}_{\mathrm{aff}}$ is binary cross-entropy on the affinity logits. $\mathcal{L}_{\mathrm{cons}}$ is the mean absolute difference between $\sigma(\ell_m)$ and M_{rec}. All loss terms are computed in fp32 outside mixed precision.

S1.4 Training and baselines

We use AdamW with learning rate 0.0003, weight decay 0.0001, cosine decay, batch size 16 and 60 epochs on 256×256 tiles, with bf16 for the forward and backward pass. The baselines use $\mathcal{L}_{\mathrm{seg}}$ on their output only. The two U-Net controls add soft clDice after the warm-up, and the second one also adds a deep-supervision head on the stride-4 decoder features with the same target and weight as our model.

S2 STRICT APLS

S2.1 Evaluator

Predictions are thresholded at 0.5 and stitched over the whole scene from the same non-overlapping 256-pixel grid as the test tiles. Ground-truth graphs on the Assam scenes come from the OSM vectors, and on the original test scenes from the label rasters. Graphs are built the same way for both. We skeletonise, prune terminal spurs shorter than 3 pixels, drop components with fewer than 9 skeleton pixels and measure edge lengths geometrically. Control points are all junctions and endpoints plus one skeleton pixel in every 9×9 cell crossed, which is about one point per 50 m. A point snaps to the nearest skeleton pixel of the other graph within the buffer, 2 pixels by default. A pair costs one if a point fails to snap or the snapped pair is disconnected, and $\min(1, |L - L'|/L)$ otherwise. APLS is the harmonic mean of the two directions. Up to 300 sources are sampled per window. Windows with fewer than two connected ground-truth control points are skipped, and at Tier 2 predicted pixels inside the ignore band are removed before vectorisation. The evaluator approximates SpaceNet in four ways, with grid-sampled control points, nearest-pixel snapping, no minimum path length and skipped empty windows. For the official numbers, the same skeletons are converted to the graph format of the official CosmiQ code and scored with its own routines.

S2.2 Results

Table S1 gives every scale and direction. Table S2 varies the snap buffer at 6 km. U-Net is slightly ahead with a 1-pixel buffer, the two models are level at 2 and 3 pixels, and our model leads at 5 pixels, so the ranking depends on how much positional error the metric forgives. Table S3 scores the Assam scenes against raster-derived

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and vector ground truth. The two agree within 0.001 for each model and give the same ranking, which supports using raster-derived graphs on the original test scenes.

S3 ADDITIONAL RESULTS

Table S4 splits the test-scene results by OSM road class, by thirds of tiles ranked by road density, and for a strict tier that scores highways only. Table S5 scores every model against the human tracings, and Table S6 gives distances between predicted and ground-truth pixels on the validation scene. Table S7 lists model size and compute, Table S8 the leave-one-scene-out folds of our model, which ran on a short schedule with uneven budgets, and Table S9 one footprint imaged on two dates. The width head reaches a mean absolute error of 0.08 pixels against 0.06 for a constant prediction of one pixel, because 97% of skeleton pixels have a half-width below 1.5 pixels.

S4 PRE-REGISTERED DECODER STUDY AND TRAINING CHANGES

S4.1 Decoder study

Before any score was computed we registered a protocol to test whether a different decoding of the same checkpoints improves strict APLS. Selection used the validation scene only, on a fixed half of its 1.5 km windows and all 6 km windows. A candidate was eligible only if its ground-truth-to-prediction score at 1.5 km and its APLS at 6 km were not below the original decoder, the primary criterion was APLS at 1.5 km, and differences below 0.002 went to the original decoder. Two amendments added candidates before their scores were viewed, and the second followed a view of the first screen. Table S10 lists every candidate for our model. Centre-line decoders recover more ground-truth routes but add spurious pieces, and no candidate beats the original fused output. For every baseline the best eligible alternative was within 0.002 of the original threshold. The frozen selection therefore kept the original decoder for every model, and the test and external evaluation equals the results in the paper. The three reconstruction variants FH, FW1 and FWC score the same as the original to the fourth decimal, which is why we describe the reconstruction branch as a training-time coupling with no measurable effect at inference.

S4.2 Training changes

A second registered protocol first screened already trained variants of our model (Table S11). A variant needed to beat the published model by 0.002 at 1.5 km without losing elsewhere, and none did. We then tried one change chosen from a validation diagnosis, a balanced region loss with equal weights on false negatives and false positives, trained as a 12-epoch continuation from the published checkpoint with a matched control that kept the original loss (Table S12). It reduced far-off predictions but also lost true routes and failed all five conditions of its gate. Finally, a wider model with 7.78 M parameters, about the size of U-Net, was trained with three seeds (Table S13). It barely changed validation strict APLS and failed its gate. Neither change is used in the paper.

Table S1: Strict APLS of our implementation across window sizes.

	APLS (symmetric)				GT→pred		pred→GT		Routes (≤25%)	
Window	Ours	U-Net	D-LinkNet	RoadFormer	Ours	U-Net	Ours	U-Net	Ours	U-Net
Test scenes (pooled)										
1.5 km	.089 ±.008	**.096** ±.005	.069 ±.009	.049	.082 ±.006	.086 ±.003	.217 ±.018	.257 ±.026	.083 ±.006	.087 ±.003
6 km	.029 ±.002	**.030** ±.003	.015 ±.003	.010	.019 ±.001	.019 ±.002	.194 ±.013	.241 ±.038	.019 ±.001	.019 ±.002
24 km†	**.008** ±<.001	**.008** ±.001	.004	.004	.004 ±<.001	.004 ±.001	.188 ±.015	.227 ±.045	.004 ±<.001	.004 ±.001
Scene†	**.002** ±.001	**.002** ±.001	.001	.001	.001 ±<.001	.001 ±.001	.200 ±.040	.244 ±.052	.000 ±<.001	.000 ±<.001
External scenes (pooled)										
1.5 km	**.103** ±.019	.095 ±.009	.079 ±.024	.051	.110 ±.021	.113 ±.025	.278 ±.029	.248 ±.017	.112 ±.022	.114 ±.026
6 km	**.041** ±.009	.037 ±.007	.025 ±.009	.015	.038 ±.010	.038 ±.014	.223 ±.022	.205 ±.019	.038 ±.010	.038 ±.014

Table S2: Strict APLS on the test scenes at 6 km for different snap buffers.

	APLS (symmetric)			GT→pred		
Buffer (px)	Ours	U-Net	Ours − U-Net	Ours	U-Net	Ours − U-Net
1	.013 ±.001	.014 ±.001	−.001 [−.001, −.000]	.008 ±.001	.009 ±.001	−.000 [−.001, +.000]
2	.029 ±.002	.030 ±.003	−.001 [−.002, +.000]	.019 ±.001	.019 ±.002	−.000 [−.001, +.001]
3	.046 ±.002	.045 ±.004	+.001 [−.001, +.003]	.030 ±.001	.029 ±.003	+.001 [+.000, +.003]
5	.072 ±.001	.066 ±.004	+.005 [+.003, +.008]	.048 ±.002	.043 ±.004	+.005 [+.003, +.007]

Table S3: Strict APLS on the Assam scenes with two ground-truth sources.

Ground truth	Ours	U-Net	D-LinkNet	RoadFormer	Ours − U-Net
Raster-derived	.103 ±.019	.095 ±.009	.079 ±.024	.051	+.008 [+.006, +.010]
OSM vectors	.102 ±.019	.096 ±.009	.079 ±.023	.052	+.007 [+.005, +.009]

Table S4: Test-scene results by road class, road density and on highways only.

	Pixel recall by OSM class			$APLS_r$ by density third			Highways only	
Model	Highways	Tert./res./uncl.	Tracks	Sparse	Middle	Dense	$APLS_r$	IoU
Ours	.564 ±.005	.348 ±.008	.122 ±.006	.335 ±.007	.265 ±.003	.185 ±.004	.568 ±.011	.044 ±.002
U-Net	.518 ±.026	.277 ±.040	.091 ±.026	.310 ±.015	.245 ±.011	.171 ±.012	.556 ±.015	.049 ±.005
D-LinkNet†	.525	.274	.091	.318	.214	.142	.547	.055
RoadFormer	.546	.303	.103	.206	.130	.108	.431	.047

Table S5: Recall on human-traced roads in 23 test tiles.

Model	Recall, traced OSM roads	Recall, untraced OSM roads	Ratio	Predictions on human-only roads
Ours	.60	.25	2.3	14.8%
U-Net	.52	.15	3.4	18.5%
D-LinkNet	.57	.23	2.5	14.7%
RoadFormer	.48	.27	1.8	11.4%

Table S6: Localisation diagnostic on the validation scene.

Model	Pred. px ≤1 px from GT	Pred. px >5 px from GT	GT px >5 px from pred.	Pred./GT length	Mask width (px)
Ours	11%	66%	70%	.80	8.3
U-Net	13%	61%	71%	.70	5.1
D-LinkNet	9%	72%	72%	.91	6.7
RoadFormer	10%	71%	70%	.94	8.2

Table S7: Model size and compute per 256×256 tile.

Model	Parameters	GMACs
Ours	3.97 M	7.8
U-Net	7.76 M	12.1
D-LinkNet	5.25 M	.95
RoadFormer	2.25 M	.77

Table S8: Leave-one-scene-out results of our model on six held-out scenes.

Held-out scene	Best epoch	IoU	F1	clDice	Road density
jun23	16	.171	.292	.236	.0682
aug23	5	.137	.241	.269	.0197
jun24	6	.123	.219	.215	.0181
oct23	7	.068	.127	.094	.0132
jan26	9	.059	.111	.096	.0066
may25	6	.002	.004	.002	.0103
Mean		.093 ±.062			

Table S9: $APLS_r$ on one footprint (path/row 101/64) imaged on two dates.

Acquisition	Ours	U-Net
23 Oct 2023 (test)	.242	.222
8 Feb 2024 (validation)	.227	.213

Table S10: Pre-registered decoder screen for our model on the validation scene.

Code	Decoder	APLS 1.5 km	APLS 6 km	GT→pred 1.5 km	pred→GT 1.5 km	Eligible
F05	fused probability ≥ 0.5 (original)	.0343	.0137	.0351	.0726	yes
M05	mask head ≥ 0.5	.0343	.0139	.0354	.0721	yes
FH	fused, reconstruction weight halved	.0342	.0138	.0351	.0722	yes
FW1	fused, reconstruction with $W \equiv 1$	.0343	.0137	.0351	.0726	no
FWC	fused, W clamped to [0.5, 1.5]	.0343	.0137	.0351	.0726	no
FS	fused and centreline ≥ 0.3	.0340	.0135	.0351	.0719	no
S03	centreline ≥ 0.3	.0292	.0177	.0474	.0415	yes
S05	centreline ≥ 0.5	.0314	.0111	.0327	.0728	no
S07	centreline ≥ 0.7	.0040	.0003	.0025	.1125	no
H1	centreline hysteresis (0.05, 0.20)	.0182	.0116	.0603	.0219	no
H2	centreline hysteresis (0.10, 0.30)	.0194	.0120	.0598	.0244	no
H3	centreline hysteresis (0.20, 0.50)	.0248	.0149	.0507	.0343	yes
BS1	original plus attached centreline pieces	.0300	.0180	.0470	.0438	yes
BS2	original plus centreline bridges	.0316	.0172	.0420	.0540	yes
BH1	original plus attached hysteresis pieces	.0198	.0120	.0597	.0251	no
BH2	original plus hysteresis bridges	.0214	.0110	.0524	.0321	no

Table S11: Screen of already trained variants of our model on the validation scene.

Configuration	Runs	APLS 1.5 km	APLS 6 km	GT→pred 1.5 km	pred→GT 1.5 km	IoU
Published model	3	.0343	.0137	.0351	.0726	.0655
− spectral stem	1	.0347	.0131	.0340	.0749	.0659
− reconstruction	3	.0350	.0136	.0346	.0786	.0684
− deep supervision	1	.0347	.0138	.0348	.0781	.0696
Anti-blob loss	1	.0324	.0111	.0329	.0805	.0615

Table S12: Balanced region loss against its matched control on the validation scene.

Run (seed 0)	APLS 1.5 km	APLS 6 km	GT→pred 1.5 km	pred→GT 1.5 km	$APLS_r$	IoU	Far >10 px	Mask width	OSM ≤2 px
Published model	.0341	.0140	.0342	.0723	.2200	.0555	.2831	8.2	.1393
Continued, original loss	.0314	.0125	.0313	.0689	.2180	.0539	.2261	8.8	.1311
Continued, balanced loss	.0304	.0109	.0291	.0728	.1961	.0529	.1733	7.6	.1180

Table S13: Wider model on the validation scene.

Model	APLS 1.5 km	APLS 6 km	GT→pred 1.5 km	$APLS_r$	IoU
Wider model (3 seeds)	.0354	.0126	.0342	.2215	.0595